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1. The waiting time (in seconds) between one song's end and the next song's beginning on a certain radio station is a random number between 3 and 6.8. Find the probability that the time between one song and another is at least 5.0 seconds.

By the definition:

A point is randomly selected from an interval (a, b).

The probability the subinterval (c, d) contains the point is defined to be $\frac{(d-c)}{(b-a)}$

$\frac{6.8-5.0}{6.8-3.0} = \frac{1.8}{3.8} \approx 47.37\%$

ANS: 47.37 %

2. Two fair 20-sided dice are rolled. How many total possible outcomes are there? What is the probability that the second number rolled is exactly 4 more than the first?

$$A = \{(1,5),(2,6),(3,7),(4,8),(5,9),(6,10),(7,11),(8,12),(9,13),$$
$$(10,14),(11,15),(12,16),(13,17),(14,18),(15,19),(16,20)\}$$

$N(S) = 20 \times 20 = 400 \qquad N(A) = 16$

$P(A) = \frac{N(A)}{N(S)} = \frac{16}{400} = 4\%$

ANS: (1) 400

(2) 4 %

3. At a certain convenience store, 40% of patrons buy hot cocoa, 45% do not buy dairy product, and 25% of patrons get both hot cocoa and dairy product. How many patrons (in percentage) get hot cocoa or dairy product?

Let A be a set of the patrons that buys hot cocoa

Let B be a set of the patrons that buys dairy product

$P(A) = 40\% \qquad P(\bar{B}) = 45\% \quad P(AB) = 25\%$

$P(A \cup B) = P(A) + P(B) - P(AB)$

$= 40\% + (100\% - 45\%) - 25\% = 70\%$

ANS: 70 %

4. Let $S = \{w_1, w_2, \ldots\}$ be the sample space of some random experiment, containing an infinite number of samples w_n, $n = 1, \ldots, \infty$. Can we define a probability distribution P on S so that $P(\{w_n\}) = \frac{1}{3^n}$? (**Hint**: You may check whether such an assignment can satisfy the Axioms of Probability.)

It is required to satisfied the P(S) = 1

$$P(S) = \sum_{i=1}^{i=\infty} \frac{1}{3^n} = \frac{a_1 \times (1 - r^n)}{1 - r} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

Which does not obey the aximos of probability, so negative

ANS: No

5. A stick of length one is broken into two pieces at a random point. What is the probability that the length of the longer piece will be at least four times the length of the shorter piece?

Let the length of the stick be 5, then the short pieces must in $(0, 1)$ or $(4, 5)$

$$P(A) = \frac{1-0}{5-0} + \frac{5-4}{5-0} = 40\%$$

ANS: 40%

6. Let $S = \{a,b,c\}$ be the sample space of a random experiment. If $P(\{a , b\})=0.6$ and $P(\{a , c\})=0.8$, find $P(\{a\})$, $P(\{b\})$, $P(\{c\})$.

$$P(\{a, b\}) = P(A) + P(B) = 0.6$$

$$P(\{a, c\}) = P(A) + P(C) = 0.8$$

$$P(\{a, b, c\}) = P(A) + P(B) + P(C) = 1$$

ANS: $P(A) = 0.4, P(B) = 0.2, P(C) = 0.4$

7. Two numbers are successively selected at random and with replacement from the set $\{1, 2, \ldots, 100\}$. What is the probability that the first one is greater than or equal to the second?

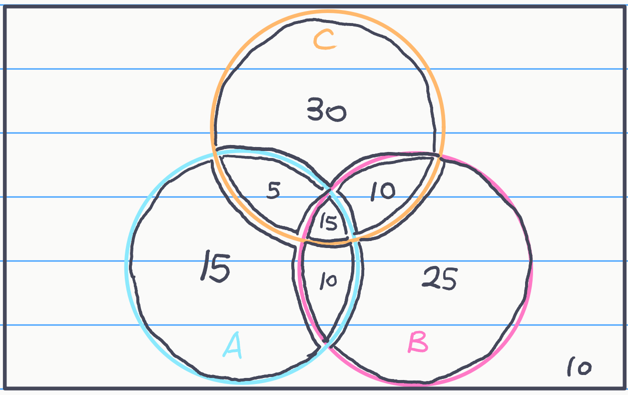
$$N(A) = \sum_{i=1}^{100} i = 5050$$

$$N(S) = 100 \times 100 = 10000$$

$$P(A) = \frac{5050}{10000} = 50.5\%$$

ANS: 50.5%

8. From a small town 120 persons were selected at random and asked the following question: Which of the three pens, *A*, *B*, or *C*, do you use? The following results were obtained: 20 use *A* and *C*, 10 use *A* and *B* but not *C*, 15 use all three, 30 use only *C*, 35 use *B* but not *C*, 25 use *B* and *C*, and 10 use none of the three. If a person is selected at random from this group, what is the probability that he or she uses (a) only *A*; (b) only *B*; (c) *A* and *B*? (Hint: You may draw a Venn diagram.)



$N(S) = 120$

$N(AB) = 25$

ANS: $P(a) = \frac{15}{120} = 12.5\%$

$P(b) = \frac{25}{120} \approx 20.83\%$

9. Use the probability axioms to prove that $P(A \cup B) = P(A) + P(B) - P(AB)$

$A \cup B = A \cup A^c B$

$P(A \cup B) = P(A) + P(A^c B)$ —— 1

$B = AB \cup A^c B$

$P(B) = P(AB) + P(A^c B)$

$P(A^c B) = P(B) - P(AB)$ —— 2

將 2 帶入 1 得: $P(A \cup B) = P(A) + P(B) - P(AB)$

ANS: 以上步驟證明

10. Let *A* and *B* be two events. Prove that $P(AB) \geq P(A) + P(B) - 1$

By probability axioms $P(A \cup B) \leq 1$

By problem 9, we have: $P(A \cup B) = P(A) + P(B) - P(AB)$

Then we can get: $P(A) + P(B) - P(AB) \leq 1$

$P(AB) \geq P(A) + P(B) - 1$

ANS: 以上步驟證明